He Wins or She Wins, You Win

Abstract

In the following paper, we take a closer look at past presidential elections in the US to see if we could infer some guidelines ahead of the results which are to be announced on November 3, 2020. The study empirically tests whether the US presidential elections do contain the important market inclusive information to explain the volatility and returns distribution of a pre-selected portfolio of stocks. We perform this event study on two portfolios, one for each National Political Party – the Democrats and the Republicans – that have been hand-picked by a major Swiss Bank. Empirical estimates conclude that the presidential elections in 2020 have a strong, significant relationship with investor's sentiment and stock market performance. Moreover, we show that election results can be capitalized to generate positive returns depending on which party takes control of the parliament. Finally, we construct two diversified portfolios – one for each political party – using quantitative techniques, that promise differentiated payoff contingent upon the respective party being voted into power. In addition, backtested performance of our strategy corroborate our results which indicate that a systematic investor can choose to bet on the elections outcome of 2020 to generate some profits at least in the short term.

I. Introduction

Political uncertainty and stock market performance remains the primary anxiety for market participants, analysts and policy makers. The stock markets' inefficiency and volatility occur due to political and economic ambiguity. In this research, we take a closer look at past presidential elections in the US and attempt to infer some guidelines ahead of the elections occurring in November 2020. Not being specialists in US Elections, we unfortunately have to restrict ourselves to observing how market participants have been influenced by the election periods and how they have expressed their preferences. The dataset is of course limited as our historical data starts in 1984 and we have only 9 elections to learn from. We look closely at two sets of portfolios that have been hand-picked by a major Swiss bank – Julius Baer. The bank claims that each portfolio is tied to a political party in the United States – Democrats and Republicans. Although, these portfolios are specifically constructed for the elections supposed to be occurring in 2020, we test the performance of portfolios that are constructed along the same guidelines as the hypothesized portfolios and conclude whether there is indeed any dependency of the portfolio performance upon which political party is voted into power in each election cycle. We not only review the performance of various "standard" equity portfolios (market, sectors, styles), but also compare the performance of the two party-linked portfolios with election betting markets and trends in polling data. We also look at whether political contributions made by corporations play a significant role in the 30 stocks being segregated into two party-linked portfolios.

We find that data from political betting markets, gathered from IOWA Electronic Markets (IEM) is a good indicator for the portfolio performance tied to each political party. While this anecdotal evidence is suggestive of the capitalization of the campaign platforms into equity prices, we acknowledge that it is difficult to separate the reaction of equity prices to this political event from other economic and financial developments. To build confidence in our model, we compare the idiosyncratic returns of each portfolio and find that the results are statistically significant. In section 2 of the paper we explain the various data sources used for our research and this event-study like framework. In section 3, we compare the two hypothesized portfolios and drill down into the differences between the two portfolios to explore factors that explain their risk and return profile. In section 4, we use various data sources to illustrate whether the hypothesized portfolios constructed by Julius Baer indeed are tied to their respective political party and whether successfully betting on a particular political party for 2020 can lead to outperformance of the respective hypothesized portfolio of stocks. In section 5, we dig deeper into the past election periods and find out whether this major political event can be used as an opportunity by investors for stock selection and generate differentiated returns at least in the short term. In section 6, we aim to build a model to construct two portfolios which are slightly different from the original portfolios yet are better tied to the respective political party. Finally, in section 7, we present our concluding remarks and takeaways from this event-study.

II. Data

Multiple data sources have been used in the following research to look at the event from various angles. We use fundamental data downloaded from Bloomberg such as stock returns, market capitalization, sector, and construct smart beta factors such as value, growth, momentum, etc. We have downloaded these scores for the S&P 500 Index constituents starting from January 1984. Utmost care has been taken to preserve Point-in-Time (PIT) data in order to avoid look-ahead bias and survivorship bias. We have used S&P 500 as a benchmark as most of the stocks in the hypothesized portfolio are a constituent of the index currently and all are large-cap stocks. The SP500 daily levels has been sourced from Yahoo Finance for the period from 1984 to 2020. The Winner Take All (WTA) data from IOWA Electronic Markets has been used as a proxy for the probability of each political party winning the elections in each period. The contract prices have been scaled to ensure that prices are between 0-1 such that they accurately measure probability of winning.

$$\pi_t^D = \frac{P_t^D}{P_t^D + P_t^R}$$

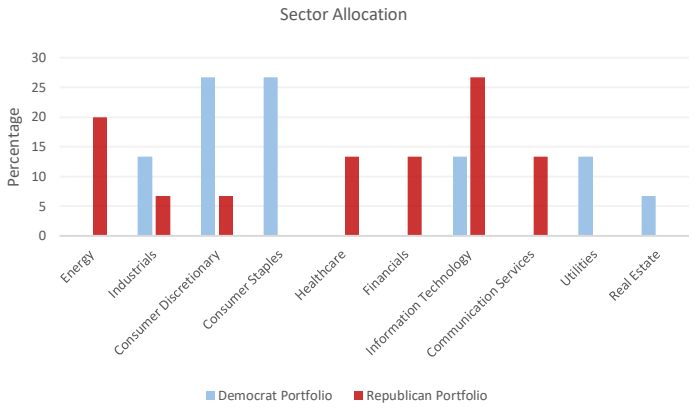
$$\pi_t^R = \frac{P_t^R}{P_t^D + P_t^R}$$

Here, π_t^D is the probability of Democrats winning at time t, π_t^R is the probability of Republicans winning at time t, P_t^D is the price of a WTA-Democrat contract at time t and P_t^R is the price of a WTA-Republican contract at time t.

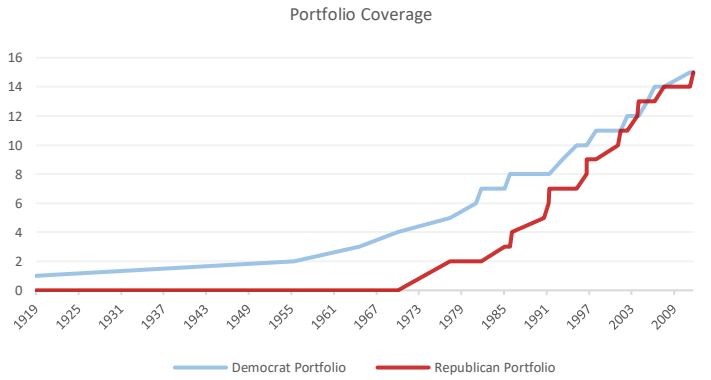
IEM has been developed by the University of IOWA and can be used as a forecasting tool. This is a real money futures market where contract payoffs will be determined by the popular vote cast in the 2020 U.S. Presidential Election. Traders can buy and sell real money contracts based on their belief about the outcome of the election. The payoff structure resembles an options contract that either pays $1 if the party receives majority of popular vote cast or 0$ otherwise. Polling data from RealClear Politics is used to track historical and latest trends in polling. Finally, the data for political contributions to each party by various companies is downloaded from the Center for Responsive Politics.

III. Are the two Portfolios Different?

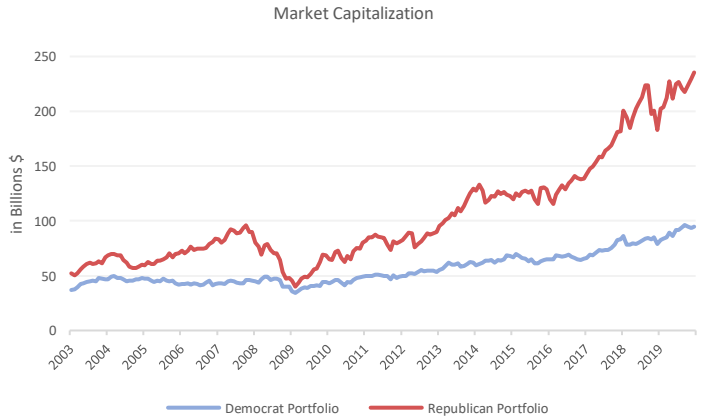
Skimming through the two portfolios hand-picked by Julius Baer tells that they are different in several ways.



Clearly, the two portfolios are invested in very different set of sectors. The Democrat Portfolio contains stocks majorly from Consumer Discretionary, Consumer Staples, Industrials and Utilities. Whereas, the Republican Portfolio contains stocks majorly from Information Technology, Energy, Healthcare, Financials, and Communication Services.



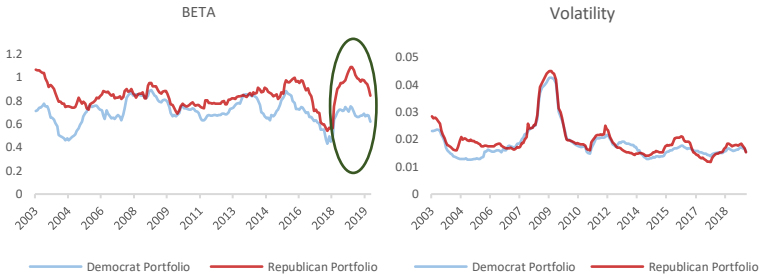
Looking at the year when each stock in the portfolio was listed as a public company, we can say that the Republican Portfolio is a much 'newer' portfolio, in that some of its stocks became public much later than their counterparts in the Democrat Portfolio. This led us to look at the two portfolios more closely and compare their performance and volatility with respect to market over time.



It can be seen that the Republican Portfolio is consistently worth more than Democrat Portfolio and that the spread has been increasing over time (here, Market Capitalization of the portfolio is calculated as an equally weighted average of the market cap of individual stocks in the portfolio). This is most likely because the Republican Portfolio contains majority of stocks from some of the biggest stocks within the Information Technology and Energy sector unlike Democrat Portfolio which contains around 50% of defensive stocks. This led us to compare the volatility and systematic risk – Beta, with respect to market – S&P500, of the two portfolios and how their risk profile has changed over time. Both volatility and beta are calculated for each stock in the portfolio using monthly returns over the last 3 years at each point in time. By performing a linear regression for excess stock returns above the risk-free rate against excess market returns, we get β for each stock along with its idiosyncratic or residual returns.

$$r_{i,t} - r_{f,t} = \alpha_{i,t} + \beta_i(r_{m,t} - r_{f,t}) + \epsilon_{i,t}$$

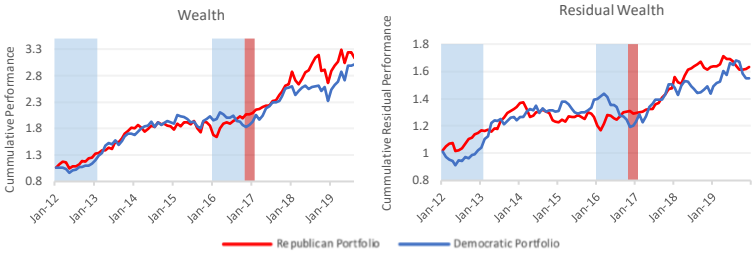
Here, r_f is the risk-free rate, ϵ_i is the residual return for the stock *i,* and r_m is the S&P 500 return at time *t.*



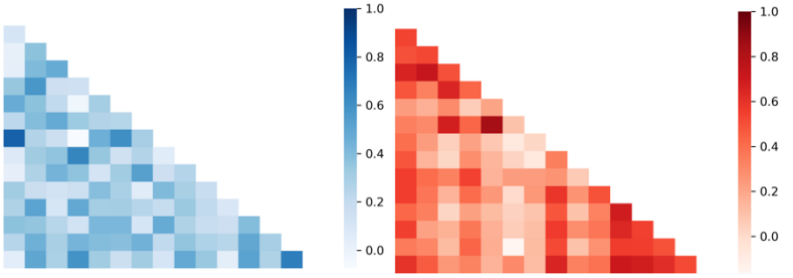
From both Beta and Volatility, it can be seen that Republican portfolio is consistently riskier than Democrat portfolio. Moreover, in the recent years, the Republican portfolio's beta has increased significantly in comparison to Democrat portfolio's beta.

We also check if there is any significant difference in the performance of both portfolios from the year 2012. We have chosen the year 2012 as the starting point to ensure that both portfolios have full coverage and to avoid missing information. Moreover, we acknowledge that the hypothesized portfolios are constructed specifically for the year 2020, so the performance of the same portfolios on a much earlier time period may not really be relevant to our research. For a more detailed explanation, we come back to this point in section IV.

It is clear that the Beta of both portfolios are significantly different, so we check cumulative performance of both portfolios using total returns and residual returns separately. Specific regimes for each term of the notes as structrued by Julius Baer have also been marked in the following graphs. The period of the notes are from Jan 29, 2020 to Jan 29, 2021. So, we have highlighted similar months in each election cycle to make a fair comparison of portfolio returns. We define a regime as the time period during which either political party was in power at the parliament.



The Republican portfolio has outperformed the Democratic Portfolio during the time period from 2012 to 2020. Looking closely, it can be seen that majority of the outperformance occurs post 2016 election regime. We comment on the possible reason for this behavior in the forthcoming sections. However, here we look at the portfolio level diversification and pairwise correlations to see if the two portfolios are different in this regard. We compute stock pairwise correlations on each month of the year 2019 within the two portfolios and then calculate average pairwise correlation during this year. Stock correlations have been calculated using last 3 years monthly returns data – for both total and residual returns. We infer interesting results from this analysis.

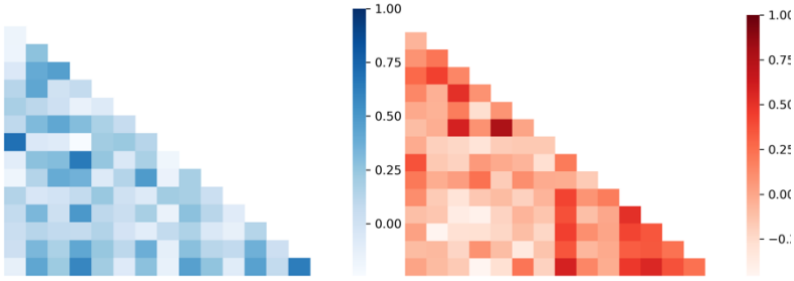


Democrat Portfolio – Total Returns

Average correlation = 29.88%

Republican Portfolio – Total Returns

Average correlation = 32.87%



Democrat Portfolio – Residual Returns

Average correlation = 14.21%

Republican Portfolio – Residual Returns

Average correlation = 2.51%

Firstly, we see that the average pairwise correlations, within the Republican Portfolio, computed using total returns for the last 3 years is higher than its counterpart Democrat Portfolio. On the contrary, if we use residual returns to compute stock pairwise correlations for the same time period, we find that the relationship is reversed. Again, this alludes to the fact that Republican Portfolio is more correlated with the market. We measure the percentage reduction achieved in portfolio volatility with Diversification Ratio of each portfolio.

$$DR_p = \frac{\sum w_i \sigma_i}{\sqrt{w^T V w}}$$

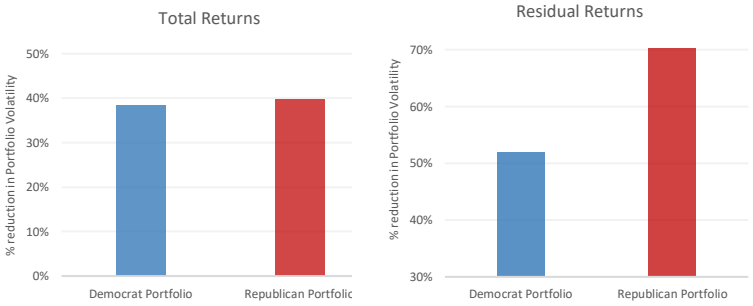
Here, V is the variance-covariance matrix of the portfolio p. In our case, since we are dealing with equal weighted portfolios, the above formula simplifies to:

$$DR_p = \frac{\sum \sigma_i}{\sqrt{V}}$$

And, % reduction in portfolio volatility is calculated as:

$$1 - \frac{1}{DR}$$

which measures the reduction in portfolio volatility achieved as compared to the weighted average of component volatilities.



The Diversification Ratio gives us similar results. For the Democrat Portfolio, DR = 2.07 while for the Republican Portfolio, DR = 3.22 in the year of 2019. In conclusion, we find that Republican Portfolio tends to be much more diversified than the Democrat Portfolio once portfolio returns have been adjusted for market sensitivity.

Political Contributions

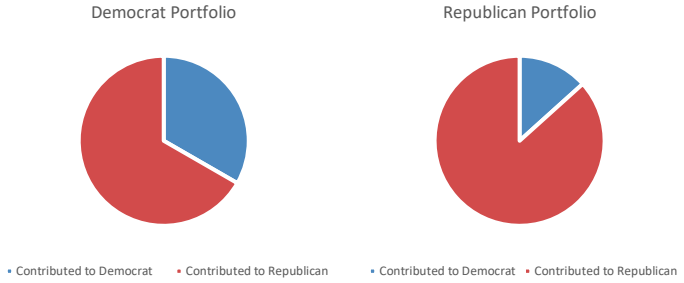
We look at the contributions made to either political party by each company in both portfolios for the two recent election cycles – 2016 and 2020. We do this to see if political contributions have any influence in the stocks being segregated into either portfolio. This data has been sourced from Centre for Responsive Politics

Average Portfolio Summary:

	Democrat Portfolio	*Republican Portfolio*
Coverage	*15*	*15*
CAGR	*14.43%*	*16.24%*
Sharpe Ratio	*1.04*	*1.15*
Maximum Drawdown	*-12.99%*	*-16.76%*
Annualised Volatility	*13.78%*	*14.70%*
Diversification Ratio	*2.07*	*3.22*
Hit Ratio	*0.68*	*0.67*

Portfolio statistics have been calculated for the period Jan 2012 to Jan 2020

which is an open source database that maintains political contributions by varioius committees and organisations for each political cycle. Our findings show that most companies have contributed to the Republican party in both election periods.

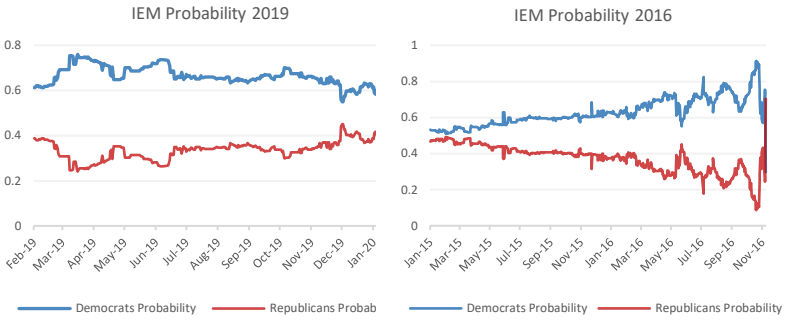


We conclude that political contributions alone are insufficient to conclude if they are the right measure to classify stocks into corresponding political portfolio.

IV. Do Elections Influence these Portfolios?

Using eveidence from the period preceeding the 2020 U.S. Presidential Election, this section attempts to test for the capitalization of election probabilities into equity prices for the sample of 30 firms hypothesized by Julius Baer. Two sources of daily data are incorporated: equity returns and candidate electoral prospects as implied by prices of political future contracts from the IOWA Electronic Market. The daily baseline estimates provide strong evidence that platforms are capitalized into equity prices. The evidence based upon weekly returns are strong suggesting a strong correlation of around 30%. To account for broader trends in equity markets during the sample period, we follow event study methodology and use abnormal/idiosyncratic returns in the analysis.

$$\epsilon_{i,t} = r_{i,t} - (\alpha_{i,t} + \beta_i \, r_{m,t})$$



We denote the price of stock i at time t by $S_{i,t}$. The elections are held at time τ and the information on the election outcome is available at time τ+1. We assume that a majority voting for the Democratic candidate would lead to an additive change in the company's value at time τ+1 of D_i per share while a Republican president would lead to a change of R_i per share. For stocks that are affected by election outcome, either $D_i > R_i$ or $R_i > D_i$. For stocks that do not react to the election outcome, $D_i = R_i = 0$. The time t risk neutral probability for a Democratic (Republican) victory is π_t^D (π_t^R). Risk neutral event outcome probabilities are inferred from political betting markets (IEM Market).

The stock price may consist of two components: one that is independent of the direct effect of the election outcome (but may include indirect effects via the market return), denoted by $S_{i,t}$. , and one that represents a direct effect denoted by $S_{i,t}^{el}$.

$$S_{i,t}^{el} = \pi_t^D D_i + \pi_t^R R_i$$
$$= (1 - \pi_t^R) D_i + \pi_t^R R_i$$

If we look at the change in Stock prices,

$$\Delta S_{i,t}^{el} = \Delta\pi_t^R (R_i - D_i)$$

Here, Δ represents differences in time interval

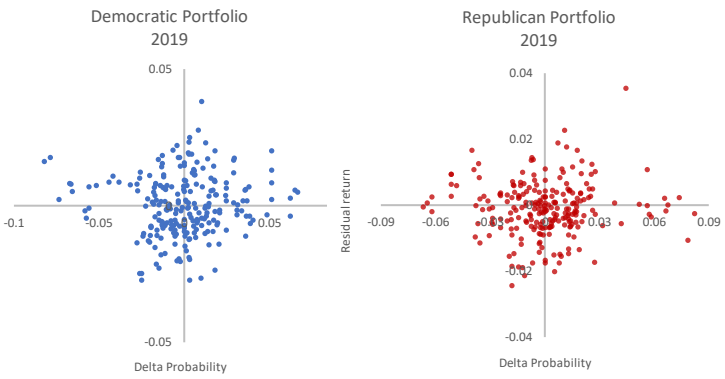
We use stock's residual/idiosyncratic returns for $\Delta S_{i,t}^{el}$ and $\Delta\pi_t^R$ is the change in probability for Republicans in the same time interval used to compute returns. Similarly, we can also write:

$$\Delta S_{i,t}^{el} = \Delta\pi_t^D (D_i - R_i)$$

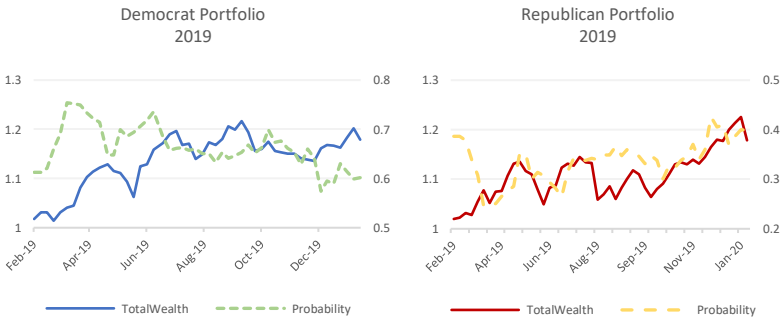
where, $\Delta\pi_t^D$ is the change in probability for Democrats. From the above equations, we note that $(R_i - D_i)$ and $(D_i - R_i)$ are stock residual returns sensitivities to the change in probabilities $\Delta\pi_t^R$ ($\Delta\pi_t^D$) for Republican and Democrats probabilities respectively.

Empirically, we see that there is a lot of noise in daily time intervals. So, we run the following regressions using weekly time intervals for both Democratic and Republican portfolios.

$$\epsilon_{i,t} = \alpha_{i,t} + \theta_i^R \Delta\pi_t^R + e_{i,t} \qquad i \epsilon R$$
$$\epsilon_{i,t} = \alpha_{i,t} + \theta_i^D \Delta\pi_t^D + e_{i,t} \qquad i \epsilon D$$



We see that there is no significant relationship between daily residual returns and change in probabilities for either portfolio. This led us to compare the cummulative performance of both portfolios with the risk neutral probability of winning of the respective party to visualise if there may be any relationship between the two at lower frequencies such as weekly or monthly.

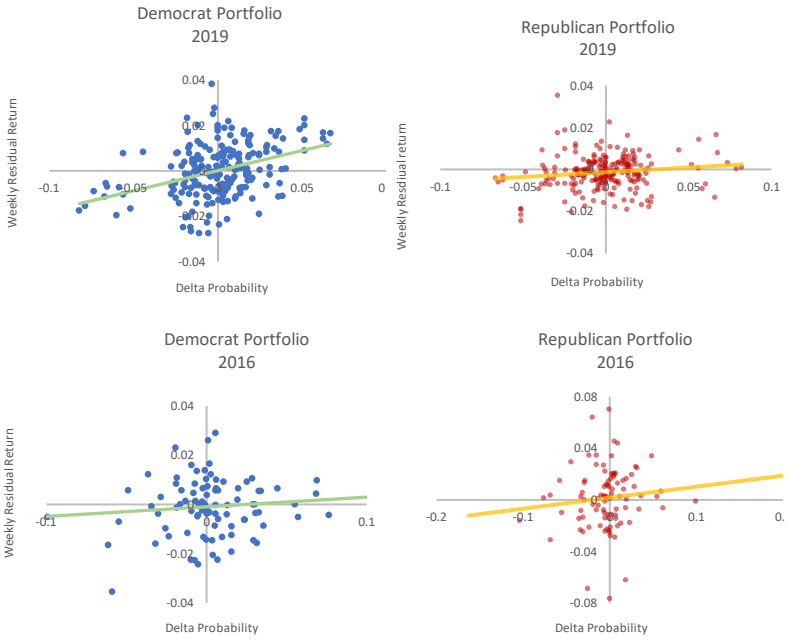


The two curves move together with correlation between cummulative performance of Democratic Portfolio and π^D = 10.62%, correlation between cummulative performance of Republican Portfolio and π^R = 8.03%.

From the equations defined above, stocks with a large positive θ_i^R are expected to benefit from a Republican victory and/or to suffer from a Democratic victory, while the opposite interpretation holds for stocks with a large θ_i^D. Stocks with θ_i^R *or* $\theta_i^D \sim 0$ should not react to the election outcome according to market expectations.

We use changes in event probabilities to validate the pre-selection, and in the forthcoming section, we use linear and non-linear machine learning techniques as the basis for selection of stocks, rendering our approach purely quantitative.

Next, we perform an OLS regression using weekly returns according to the equations defined above for 3 different portfolios – Democratic, Republican, and a long/short portfolio which goes long on the Democratic Portfolio and goes short on the Republican Portfolio. The results are quite interesting and are evidence of a strong linear relationship between weekly changes in probabilities measured from election betting markets vs weekly residual returns. We do the same for the election cycle of 2016 as well to check if the results are persistent and significant.



It is seen that the relationship within the Democratic Portfolio is stronger and more significant than the Republican Portfolio. This implies that average weekly abnormal returns of the democratic portfolio are explained more by the betting markets data than its counterpart republican portfolio, i.e. a lot of the returns within the republican portfolio is still unexplained even after accounting for effects of election in the portfolio.

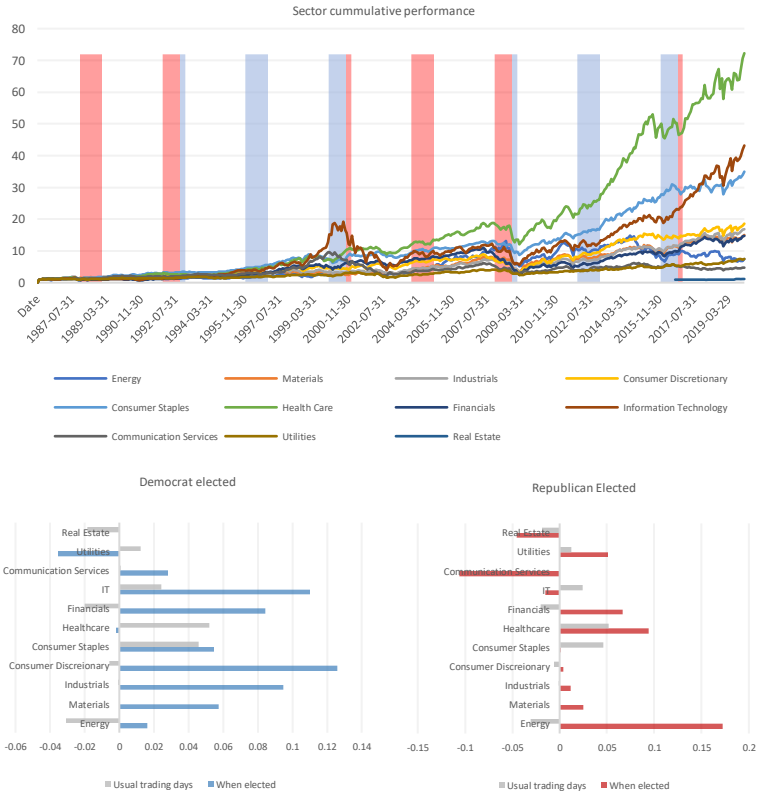
For the year 2016, we see that although the results are in line with the expected theoretical equation, but they are not significant. This implies that the hypothesized portfolios may be well suited for the 2020 elections but not for 2016 elections. The R_2 for the Democratic portfolio is around 15% while that for the Republican portfolio is merely 2.1%. We interpret this result by saying that the Democrat portfolio is a better tied to the Democratic party than the Republican portfolio is to the Republican party. The low R_2 reflects that while the electoral measures do not fully explain the variation in equity returns, news is incorporated into equity prices, and that this relationship is economically and statistically significant at least in the short term.

	Coeff $\Delta\pi$	*Intercept*	*t-stat $\Delta\pi$*	*Significant*	*R_2*
Dem_2016	0.041	-0.001	0.893	No	0.012
Rep_2016	0.0852	0.001	1.005	No	0.011
Dem_2019	0.1773	-0.0003	6.072	Yes	0.1512
Rep_2019	0.0452	-0.0013	2.005	Yes	0.021
Dem_Rep_2016	0.0451	-0.002	1.437	No	0.0224
Dem_Rep_2019	0.066	0	2.031	Yes	0.024

Regression stats for the hypothesized Democratic, Republican and long/short Democratic-Republican portfolio for the years 2015-2016 and 2019-2020.

Sector Analysis

We aim to check how various sectors (according to GICS classification) have performed during each election cycle in the past. We maintain our investment period from January of election year to January next year to be consistent with the investment horizon of the notes constructed by Julius Baer. A sector's return is calculated as the average return of all the stocks within the sector in S&P 500 at that time.



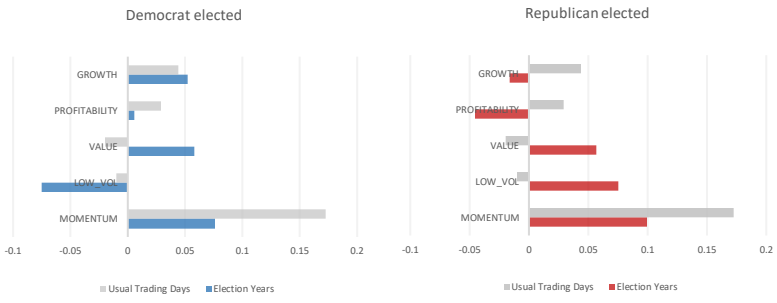
*Here, **when elected** is the January of election year to January of the next year when the party is elected, consistent with our investment horizon of the structured note. Usual trading days are all the days outside of our investoment horizon.*

The sector returns comparison between usual trading days and when elected, from the year 1987, is done after adjusting for other macro effects which we assume is priced into the market. Our results indicate that election of a Democrat President is most favorable to sectors such as Consumer Discretionary, Industrials, Information Technology, and Consumer Staples. Looking at the sector allocation of the hypothesized Democratic Portfolio, we find that roughly 80% of the portfolio is invested in these 4 sectors.

The election of a Republican President is most favorable to sectors such as Energy, Healthcare, Financials, and Utilities. Looking at sector allocation of the hypothesized Republican portfolio, we find that roughly 50% of the portfolio is invested in these 4 sectors. An interesting difference to note is that IT and Communication Services hold up to 40% of the Republican portfolio while our analysis tells us that these two were one of the worst performing sectors during the investment period of our interest, when a Republican president is elected.

Style Analysis

Next, we repeat the same event analysis on the 5 styles that we have created – Value, Growth, Momentum, Low Volatility, Profitability. We measure Value by the trailing 12 months Earnings Yield, Growth by expected EPS growth, Momentum by last 12 months returns less last 1 month return, Low Volatility as inverse of realized volatility in the last 3 years calculated using monthly returns, and Profitability by Return on Equity.



Similar to sector returns, style returns have been calculated using residual returns to remove market effects. For each style, stocks within SP500 are ranked in decile (1-10) on a month *t*. A long/short portfolio is constructed which goes long on the top decile and goes short on the bottom decile. For this L/S portfolio average residual returns are calculated from the month *t* to *t*+1. This long/short return calculated is taken to be style return for the month *t*+1. This process is repeated for every month to get the style's return distribution.

We note that a Democratic elected president is favorable to Growth and Value stocks and is unfavorable to Low_Vol stocks. On the contrary, a Republican elected president is favorable to Value and Low_Vol stocks and unfavorable to Profitability and Growth stocks. Interestingly, we find that Value stocks perform better in both election regimes than usual trading days and Momentum stocks underperform in both election regimes than usual trading days.

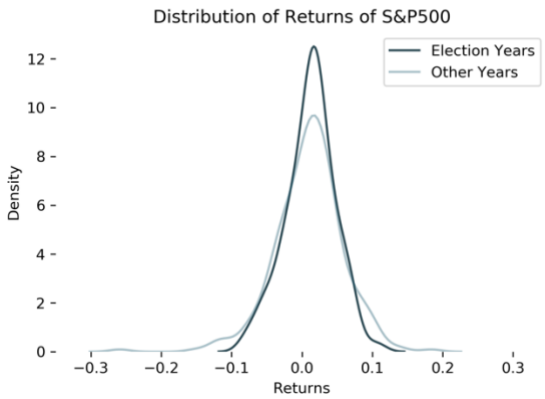
The above analysis enables us to deduce that both Democratic and Republican portfolios structured by Julius Baer show modest relationship with characteristics that have outperformed in the past election years. Although, the relationship is stronger in case of Democratic portfolio than Republican portfolio.

V. Is there an effect of elections on asset returns?

From 1984 there have been 9 elections. As shown below, the average monthly return of the S&P 500 index during the election years was 1.19% (annualized return = 14.28%) while the average monthly return during normal trading days was 0.98% (annualized return = 11.76%). In calculating average returns during elections, we have ommited the year 2008 considering the year as outlier. Our assumption is that in that year, elections must have had little effect in the market performance because of the recession and its influence on other macro indicators. We perform the Welch's t-test as a stististical significance test to check if the mean of the two distribution are equal.

$$t = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

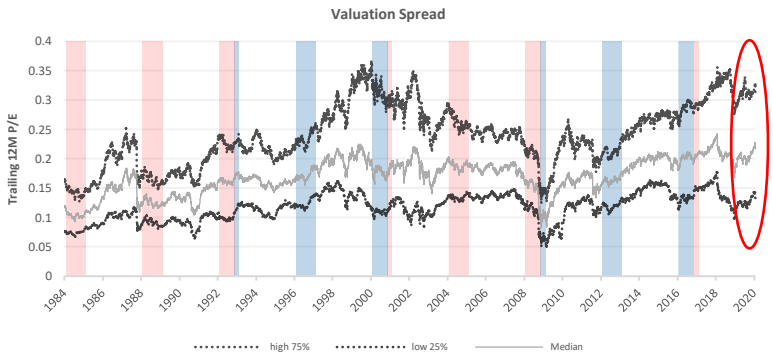
Here, $\overline{X_1}$ and $\overline{X_2}$ are the sample means of the two distribution, $s_1$and s_2 are the sample standard deviations, N_1 and N_2 are the number of samples in each distributions.



Running this test, we get a t-statistic of 0.4654 and a corresponding p-value of 0.643. Thus, we fail to reject our null hypothesis that the means of the two distribution are equal, at the 5% level of significance. So, we dig deeper to look at other metrics for the effect of elections on the market as a whole.

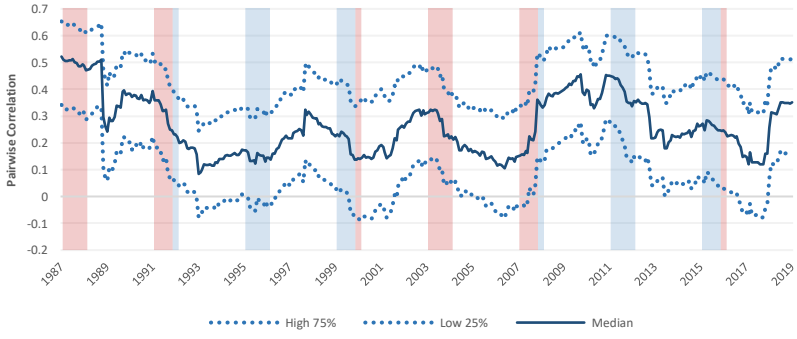
Opportunity Set

Another metric we look closely is the "opportunity-set" for investors. We think of this as the available opportunity for stock selection that investors can pursue to generate differentiated reutrns. We measure opportunity set in two different ways - valuation spread and stock pairwise correlation. Widening valuation spread typically indicates more stock level differentiation and therefore a better environment for stock selection. On the other hand, narrowing valuation spreads are indicative of lower levels of stock differentiation. The figure below shows the median, 25th percentile, and 75th percentile of trailing price to earnings for the S&P500 index constituents.



We see that the spread is increasing than prior months for most of the election periods and also for the current election period. This suggests that there may be some opportunity that investors can exploit to generate differentiated portfolio returns.

Closely related to the valuation spreads is the median pairwise correlation among stocks in the market. This is calculated by taking every possible pair of stocks and computing the correlation of their monthly returns based on the past 24 months of data and then taking the median across all those pairs.



It is easy to see that average pairwise correlations tend to decrease during all election years (except the year 2008 which we have considered as an outlier). This tells us that during the year of the election, stocks tend to trade more on their own merits, rather than being driven by common factors.

In conclusion, we find mixed results by looking at the reutrn and risk distribution of stocks during the year of the election versus other trading days. Nevertheless, since stock correlations decrease, we believe that investors may utilize this opportunity to create diversified portfolios while maintaining similar levels of expected returns to reduce the risk of their overall portfolio considerably.

VI. Portfolio Construction

In the final section we utilise all the results learnt so far to construct two long-only portfolios – Democratic and Republican portfolio. Each portfolio is constructed such that they are expected to outperform the market, contingent upon the election results – respective political party being elected into the parliament. Thus, we believe that the portfolios constructed can be used as a proxy for investors who are interested to bet on the outcome of U.S. Presidential Elections 2020. This assumption is in line with that of the construction of structured notes by Julius Baer that they intend to sell to institutional clients with similar interests. We also use state of the art quantitative and machine learning techniques to construct the two portfolios.

In our approach, we assume that the stocks are selected from the universe S&P 500. This is because the hypothesized portfolios constructed by Julius Baer are all large-cap stocks and 28 out of the 30 stocks are constituents of the S&P500 index. Secondly, both portfolios are U.S. large-cap long-only that contain 15 stocks in each portfolio. Again, this assumption is in line with the way that Julius Baer has constructed their structured notes.

The model incorporates a number of unique features including dynamic factor selection, a non-linear tree component, a linear classification model, and active style and sector rotation. Besides these, we have also included the factor that we have created using data from Iowa Electronic Markets. This 'alternative' factor measures the past correlation between stock's residual return with the probability of winning of a political party captured from political betting markets.

We have formulated the exercise as a supervised learning classification problem. We develop two models – one for each political party - using similar techniques. Each model is meant to select 15 stocks that are expected to outperform the most in the market contingent upon the respective political party being elected to form the parliament. In case of a conflict i.e. if a stock is predicted to be in both the portfolios, we put the stock in the portfolio where it is expected to outperform more based on its probability of outperformance. The target variable for the model is binary where 1 represents market outperformance between the period January of election year to January of next year. This is in line with the investment horizon of the structured note created by Julius Baer.

Our model is an ensemble of two different machine learning algorithms – 1. Logistic Regression that captures linear relationship between predictors and target (market outperformance) and, 2. XGBoost or Extreme Gradient Boosting that captures complex non-linear relationships between predictors and target. Finally, we use the average of the predicted probability from both algorithms as our final criteria for selecting stocks within the portfolios.

Logistic Regression

As mentioned in the previous section, one of the biggest challenges in this exercise is the limited number of election events. The small sample size poses significant difficulties on statistical inference.

$$logit(P) = log\left(\frac{P}{1-P}\right) = \beta_0 + \beta X^T$$

Here, P is the probability that a stock outperforms the market, β is the vector of weights as learned by the logit model for each predictor, and X^T is the transpose of a vector of predictors fed into the model.

XGBoost

The gradient boosting classifier is an additive model using classification trees of a fixed size as weak learners $h_m(x)$. Thus, we can denote our model $F(x)$ as a weighted sum of M weak learners.

$$F_M(x) = \sum_{m=1}^{M} \gamma_w \, h_m(X)$$

When the m_{th} learner is added to the model, it becomes

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

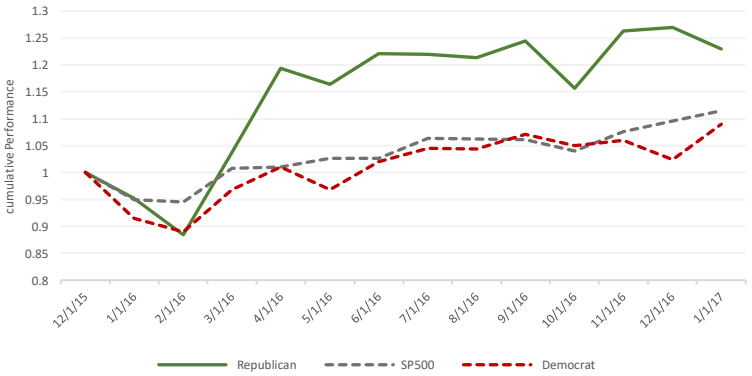
Results and recommendations

We trained our model using data from election years starting from 1984 and stocks that outperformed in those years and used the models to predict stocks that are expected to outperfrom in the 2020 elections. We get the following 15 stocks in each portfolio:

Democratic Portfolio	Republican Portfolio
Brighthouse Financial Inc	The Kraft Heinz Co
Allergan PL	Molson Coors Beverage Co
Hewlett Packard Enterprise Co	DENTSPLY SIRONA Inc
CenturyLink Inc	Facebook Inc.
News Corp	Coty Inc
Micron Technology Inc.	Kinder Morgan Inc.
Nektar Therapeutics	Incyte Corp
Xerox Holdings Corp	Alphabet Inc.
Western Digital Corp	McKesson Corp
Qorvo Inc	Cardinal Health Inc
Mylan NV	Zimmer Biomet Holdings Inc
DXC Technology Co	J.M. Smucker Co (The)
Perrigo Co Plc	Amazon.com Inc
Corning Inc	CVS Health Corp
Walmart Inc	Baker Hughes a GE Co

Backtest performance

We test the performance of our model on the election year 2016. To do this, we train our model using all the data before January 2015 and predict using the data from January 2015 to January 2016. Since Republicans were voted to form the government in 2016, we look at the performance of the republican portfolio, between January 2016 to January 2017, as predicted by our model.



The portfolio that our model predicted, generated an annualized return of 22.97% compared to 11.5% of S&P 500 index and has considerably outperformed the Democrat portfolio. This is strong evidence that the model has selected desirable portfolios that are more fine tuned bet on the result of the elections in 2016.

VII. Conclusion

Using evidence from past Presidential Elections, this paper has demonstrated that information from election betting markets are capitalized into equity prices for a sample of 30 firms in the US that are hypothesized to be tied to the political parties. To study investor preferences, we have looked at how characteristics from polling and political betting markets translate into returns for related stocks. We observe that we can find some of the relative valuation (i.e. the premia or discount applied by investors) by tracking correlation between stocks' abnormal returns with probability of winning of political party that the stocks are tied to. These correlations explain over 15% in the variation of the Democrat's portfolio's average residual returns; however, the relationship is not as strong in the case of Republican portfolio.

Looking at past elections and the way markets have performed in each regime, we found that the Democrats are most favorable to sectors such as Consumer Discretionary, Consumer Staples, IT, and Industrials while the Republicans are most favorable to sectors such as Energy, Healthcare, Financials, and Utilities. Thus, in case of anti-incumbency in the year 2020, there may be sufficient opportunity posed by sector rotation. On the contrary, in case of Republicans being re-elected, investors can capitalize by a higher exposer to styles such as Momentum, Value and Low Volatility.

We also noticed that elections have an effect on how the market performs at least in the short term. The average annualized return of the SP500 index during election years (January of election year to January of next year) is 14.28% while average annualized return during usual days in 11.26%. We see that stock pairwise correlations tend to decrease during election year which implies that there may be some opportunity for investors to construct portfolios with an aim to reduce their overall risk.

Finally, we used an ensemble machine learning approach to construct two long-only portfolios that are expected to be better tied to each of the two political parties. We suggest the portfolios predicted by the model with a belief that those portfolios better capture the relationship with respective political party than the hypothesized portfolio selected by Julius Baer. Backtest performance of the model on the year 2016, shows that our model picked a Republican portfolio that generated close to 23% returns during the period January 2016 to 2017 compared to 11.5% of the S&P 500 index and 9% of the Democratic portfolio selected by our model.

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